Bardia Sabagh Kermani

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RESEARCH INTERESTS

Computer Vision
Efficient & On-Device Deep Learning
Graph Neural Networks
Datasets & Benchmarking

EDUCATION

B.S. in Computer Engineering. 2022 – Feb 2027 (expected)
Amirkabir University of Technology, Tehran, Iran.
- CGPA: 17.56 / 20 (3.64 / 4.0) • Last 64 Credits: **18.64 / 20 (3.97 / 4.0)**

RESEARCH EXPERIENCE

- Research Assistant, Data Science Lab 2026 – Present
  - Advisor: *Dr. Mostafa Chehreghani*
  - Researching LLM integration into GNNs and knowledge graphs
  - Conducting an ablation study of TxGNN (*Nature Medicine*, 2024), testing whether its zero-shot drug-repurposing gains stem from prior clinical knowledge rather than learned GNN representations (comment paper in preparation)
- Research Assistant, Intelligent Computing & Communication Infrastructure Lab (ICCI) 2025 – 2026
  - Advisor: *Dr. Hamed Farbeh*
  - Real-time, on-device obstacle detection for pedestrian safety; three manuscripts under review, all available on arXiv (listed below)

Manuscripts Under Review preprints on arXiv

- A. Zandi, S. Nami, R. Ganjipour, **B. Sabagh Kermani**, A. Tarabkhah, H. Farbeh. *Real-Time Obstacle Detection and Distance Estimation on Smartphones for Visually Impaired and Distracted Pedestrians.* Submitted to IET Cyber-Physical Systems: Theory & Applications. *arXiv* | *code*
  - *My role:* contributed to implementing the early-exit heads, and implemented ARCore depth-based distance estimation with a monocular focal-length fallback and the on-device benchmarks; the compressed YOLOv8n ran up to 8× faster on a mid-range phone (3.2 s → 0.39 s), recommended setup at 0.42–0.48 s with 0.81 recall.
- A. Zandi, A. Azami, **B. Sabagh Kermani**, P. Abbasian, R. Ganjipour, S. Nami, H. Farbeh. *ROD-Dataset: A Pedestrian-Perspective Benchmark of 25 Urban Obstacle Classes for YOLO-Based Detection.* Submitted to Multimedia Tools and Applications. *arXiv* | *Kaggle* | *HuggingFace*
  - **24,326 images** and **40,195 annotated instances** across **25 classes**, merging 29 public collections with street-level imagery captured in Tehran and Toronto; *My role:* collected and annotated part of the field-captured imagery.
  - Benchmarked six YOLO generations (v8–v26), showing recall and mAP rankings disagree; open source (MIT) with **88K+ Hugging Face downloads**.
- A. Zandi, A. Ashrafi, S. Nami, S. S. Neghahban, A. Entezari, P. Abbasian, A. Azami, P. Taherzadeh, **B. Sabagh Kermani**, R. Ganjipour, H. Farbeh. *From White Canes to Edge AI: A Survey of Assistive Navigation Technologies for Blind and Low-Vision Individuals.* Submitted to Disability and Rehabilitation: Assistive Technology. *arXiv*
  - PRISMA-based review of 108 studies (2008–2026) in a four-era taxonomy; contributed the literature review of lightweight YOLO methods (pruning, quantization, attention) for its edge-optimization synthesis.

SKILLS & EXPERTISE

- **Computer Vision & Deep Learning:** PyTorch, Ultralytics/YOLO, Object Detection, Pose Estimation, Model Compression, PyG
- **Programming Languages:** Python, Kotlin, Java, C/C++
- **Technologies & Tools:** Docker, Prometheus, Grafana, Git, LaTeX

WORK EXPERIENCE

Backend & AI Platform Engineer Nov 2024 – Nov 2025
Shatel AI, Tehran

- Co-architected and deployed production RAG chatbot platforms (vector retrieval + LLM inference) for Sales, Support, and Lead Generation.
- Built high-concurrency API infrastructure (REST/gRPC/WebSocket, Redis, PostgreSQL) with full observability of latency and AI service health.

NOTABLE PROJECT

- **IO Motion: On-Device Pose Estimation & Fitness Analysis** **GitHub**
  Android app (Kotlin, MediaPipe Pose Landmarker) analyzing squats, sit-ups, push-ups, and planks from live camera or video – rep counting, biomechanical KPI scoring, fully offline

TEACHING ASSISTANTSHIPS

- Data Mining (Head TA) Fall '25, Spring '26
  - Instructor: *Dr. Ehsan Nazerfard*
  - Led TA team; designed & graded HWs, quizzes, and exams
  - Conducted supplementary teaching sessions
- Microprocessor and Assembly Language Fall '25
  - Instructor: *Dr. Hamed Farbeh*
  - Designed course project and graded student submissions
- Algorithm Design / Data Structures & Algorithms Fall '25, Spring '25
  - Instructor: *Dr. Alireza Bagheri*
  - Prepared and graded homework assignments

COURSES & CERTIFICATIONS

- Online: CS231N: Deep Learning for Computer Vision Stanford – *Fei-Fei Li*
- Online: CS236: Deep Generative Models Stanford – *Stefano Ermon*
- Online: CS224W: Machine Learning with Graphs Stanford – *Jure Leskovec*
- Big Data Analysis (Graduate Course; Grade: **18/20**) AUT – *Dr. Mostafa Chehreghani*

HONORS & AWARDS

- Placed 6th in the *Fito Hackathon*, a national Data Science & Image Processing competition.
- Ranked among the top 10 contestants in the *AlgoCup* coding competition.

SPOKEN LANGUAGES

- English (**TOEFL iBT 5.5/6.0;** 110/120 on the previous scale)
- Persian (native)

Last Update: October 6, 2026